GOVERNMENT COLLEGE OF ENGINEERING, NAGPUR

Department of Computer Science & Engineering

Nagpur, Maharashtra - 441108

COMMUNITY ENGAGEMENT PROJECT

PROJECT SYNOPSIS

NIRMAN DATA AUTOMATION PLATFORM

A Dynamic Form-Aware Batch Data Extraction and Analysis Platform

Submitted to

Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur

in partial fulfillment of the requirements of the Community Engagement Project

SUBMITTED BY

Uday Dhote	Enrollment No.: _____
Tejas Jadhao	Enrollment No.: _____
Vanshaj Dudhe	Enrollment No.: _____

Under the Guidance of

Dr. Pushpanjali Chouragade

Academic Year 2026–2027

Date of Submission: _____

1. ABSTRACT

NIRMAN Data Automation Platform is a form-aware batch data extraction and analysis system designed to convert information from handwritten and printed forms into structured digital records. It integrates Google Gemini Vision AI with a schema-driven workflow, field-level validation, batch processing, data-quality controls, Excel reporting, descriptive analytics and optional AI-generated analytical reports. A representative form is first analyzed to propose a schema; the user reviews and confirms the fields and constraints before the batch is processed. This schema then guides extraction and deterministic validation. The platform supports rating, numeric, text, categorical, checkbox, date and other fields. Raw AI output is kept separate from validated values, while explicit status values preserve missing, illegible, invalid, out-of-range, API and parsing failures. The resulting four-sheet workbook provides data records, field definitions, data-quality information and summary statistics. The project connects software engineering and AI integration to a practical community need: reducing repetitive form transcription while keeping uncertainty visible and human review available.

Keywords: form digitization, multimodal AI, Gemini Vision, schema-driven extraction, data validation, Flask, Pydantic, Excel automation, descriptive analytics, human-in-the-loop.

2. INTRODUCTION AND BACKGROUND

Workshops, training activities and community programmes often collect participant feedback using paper forms. Paper-based collection is convenient where participants may not have individual digital devices or reliable connectivity, but the information must later be digitized before it can be aggregated, analyzed or used for reporting. Manual transcription is repetitive, takes staff time and can introduce mistakes, particularly when handwriting is unclear or forms contain many fields.

NIRMAN addresses the post-collection stage by combining visual AI interpretation with software-controlled validation and familiar spreadsheet outputs. It is designed not only to extract text, but also to maintain a description of expected fields, check values against constraints, expose uncertain records to the operator and organize processed data for analysis.

3. PROBLEM STATEMENT

Organizations that collect handwritten or printed forms face a time-consuming process of reading responses, entering them into spreadsheets, checking data quality and preparing summaries. A dependable solution must process forms in batches, adapt to different field structures, recognize when a value is missing or uncertain, validate extracted information and produce outputs that can be reviewed before being used for decisions.

4. PROJECT AIM

To provide a reusable, AI-assisted platform that digitizes batches of handwritten and printed forms and converts them into structured, validated records and analytical outputs, while retaining human oversight for uncertain or invalid values.

5. OBJECTIVES

- Automate extraction of structured information from handwritten and printed forms.
- Generate an editable data schema from a representative sample form instead of assuming a fixed list of questions.
- Support rating, numeric, text, categorical, checkbox, date and other field representations.
- Validate extracted values against field-specific types, ranges, required status and allowed-value constraints.
- Keep raw AI output distinct from validated values and status information.
- Flag illegible, missing, invalid, out-of-range, API and parsing failures for appropriate review.
- Enable batch upload, progress visibility and correction of extracted values in the web interface.
- Generate a four-sheet Excel workbook containing records, definitions, data quality and summary statistics.
- Provide descriptive analytics and an optional plain-language analytical report.
- Apply practical file-handling, credential-protection and audit-logging controls.

6. SCOPE OF THE PROJECT

The project covers the workflow from a sample form to reviewed digital output. It includes schema proposal and editing, schema persistence, image upload, batch extraction, field-level validation, exception tracking, manual correction, Excel generation and optional report generation. The documented input formats include JPG, PNG, TIFF, BMP and WEBP images.

The platform is intended for structured forms whose fields can be described by the supported schema. It does not guarantee that every handwritten response will be recognized correctly, nor does it replace human judgement for ambiguous cases. Accuracy, throughput, time saved and financial benefit must be measured using an appropriately verified evaluation dataset rather than assumed.

7. TARGET USERS AND APPLICATION AREAS

Potential user group	Possible use
Educational institutions	Workshop feedback, course evaluations, student surveys and event forms.
Community and nonprofit organizations	Participant feedback, programme evaluation and outreach forms.
Administrative teams	Registration, application and structured paper records.
Research and survey teams	Questionnaire digitization, response summaries and quality checks.

These are potential application areas based on the platform's functionality, not claims of confirmed customer adoption or deployments.

8. SIGNIFICANCE FOR COMMUNITY ENGAGEMENT

The community-oriented value of NIRMAN is its ability to make collected feedback easier to digitize, review and summarize. Where paper remains a practical collection method, the platform can reduce clerical effort after collection and make data-quality problems more visible. It is an assistive tool: it supports analysis but does not make the organization’s decisions or remove the need for responsible handling of participants’ information.

9. PROPOSED METHODOLOGY AND WORKFLOW

The platform uses a five-step workflow in which a human confirms the form definition before batch processing. This reduces the risk that a mistaken field interpretation will silently propagate across every record.

1. Analyze	2. Review	3. Upload	4. Process	5. Report
Upload one representative form	Edit and confirm schema	Select the form batch	Extract, validate and review	Export workbook and optional report

9.1 Step 1 - Analyze a Sample Form

The user uploads one representative image. The sample analyzer uses the vision model to identify likely fields, field types and constraints, then proposes a candidate schema. The proposal is a starting point for review, not an unquestionable source of truth.

9.2 Step 2 - Review and Confirm the Schema

The user checks field labels, expected types, required status, numeric bounds, allowed categorical values and date formats. Fields can be edited, added or removed, and the confirmed schema can be saved as JSON for reuse in recurring workflows.

9.3 Step 3 - Upload the Batch

The user uploads the forms that share the confirmed schema. File handling checks permitted extensions and upload limits, sanitizes filenames and handles duplicate filenames without intentionally overwriting existing inputs.

9.4 Step 4 - Extract, Validate and Review

The extraction engine supplies each image and the confirmed schema to the AI provider using structured instructions. A two-pass workflow first identifies potentially messy fields and then extracts values. The returned output is parsed, represented as structured field values and checked by deterministic validation rules. Records with uncertain or invalid values receive explicit statuses and can be corrected in the browser interface.

9.5 Step 5 - Export and Report

Validated records feed the Excel exporter and analytics module. The optional report generator produces a plain-language summary from the same processed dataset. The operator should inspect data-quality results before interpreting or sharing the output.

10. SYSTEM ARCHITECTURE AND DESIGN

The architecture separates the browser interface, Flask application controller, core processing modules, external AI provider and output/storage layers. This separation keeps visual interpretation distinct from validation and reporting, allowing each component to have a clear responsibility.

Layer	Responsibility
User interface	Five-step wizard, upload controls, progress display, record review and manual editing.
Flask web application	HTTP routes and coordination of sessions, schema operations, batch status and record review.
Core NIRMAN modules	Schema manager, typed models, extraction, validation, analytics and reporting.
External services and outputs	Gemini Vision API, file storage, JSON schemas and Excel workbook generation.

11. PRINCIPAL MODULES

Module	Main responsibility
config.py	Central configuration, file paths, runtime limits and status definitions.
models.py	Typed representations of schemas, field values, extraction records and analytics.
ai_provider.py	Gemini API wrapper, retry handling and rate limiting.
sample_analyzer.py and schema_manager.py	Candidate schema generation, editing support and persistent schema operations.
extractor.py and validator.py	Structured two-pass extraction and type-aware validation.
excel_export.py and analytics.py	Four-sheet workbook generation, descriptive statistics and frequency distributions.
report_generator.py	Optional plain-language analytical report.
file_handler.py and audit.py	Upload validation, safe filenames and privacy-aware operational logging.
app.py and templates	Flask routes and the user-facing wizard.

12. TECHNOLOGY STACK

Technology	Use	Role in the platform
Python	Application logic	Coordinates extraction, validation, analytics and exports.
Flask	Web framework	Provides the web application and API routes.
Google Gemini Vision API	AI service	Interprets images and returns candidate field values.
Pydantic	Data models	Provides structured models for schemas and records.
JSON	Schema persistence	Stores form definitions for reuse.
HTML / JavaScript	Web interface	Supports the guided workflow and user interactions.
Excel generation libraries	Reporting	Creates the structured workbook.
pytest	Testing	Supports automated tests for deterministic modules.

13. DATA VALIDATION AND ERROR HANDLING

AI-based extraction can be uncertain; deterministic validation checks whether an extracted value complies with the confirmed schema. NIRMAN keeps the raw AI value separate from its validated value and status. This allows the operator to distinguish model output from what the application accepts for downstream processing.

Status	Meaning	Handling principle
ILLEGIBLE	The response cannot be read confidently.	Keep visible and request human review.
FIELD_MISSING	A required response is blank or absent.	Do not confuse with unreadable content.
OUT_OF_RANGE	A numeric value violates configured bounds.	Do not treat it as valid numeric data.
INVALID_VALUE	The value violates its type or allowed set.	Flag for correction.
API_ERROR	AI service request failed.	Preserve the failure state.
PARSE_ERROR	Returned output cannot be parsed as expected.	Prevent malformed data from being accepted.

14. OUTPUTS AND ANALYTICS

- **Data Records:** one record per form, extracted values and review flags.
- **Field Definitions:** field names, labels, types, constraints and groups.
- **Data Quality:** valid, missing, illegible and invalid counts, plus response rates.
- **Summary Statistics:** mean, median, standard deviation, minimum, maximum, mode and frequency distributions where applicable.
- **Optional report:** plain-language summary generated from the processed dataset.

15. SECURITY, PRIVACY AND RESPONSIBLE USE

The documented safeguards include environment-based API credential configuration, upload-extension and file-size checks, filename sanitization, duplicate-file handling, path-traversal protection and privacy-aware audit logging. These are practical safeguards but do not constitute a complete security certification. Before processing personal data, the operator should establish appropriate access restrictions, retention/deletion rules and clear handling practices for information sent to an external AI service.

16. TESTING AND EVALUATION

Project documentation records 85 automated tests across validation, models, analytics, Excel export, schema management, report generation and file handling. These tests support regression checking of deterministic parts of the application. They do not by themselves establish real-world handwriting-recognition accuracy.

A formal evaluation should compare extracted values with manually verified reference forms and report field-level exact-match accuracy, rating accuracy, exception detection, false-flag rate, human correction rate, processing time per form and cost per form. No measured value should be claimed until the benchmark has been performed.

17. EXPECTED OUTCOMES AND COMMUNITY BENEFITS

- Reduced repetitive transcription effort for batches of paper forms.
- More consistent structure through confirmed schemas and type-aware validation.
- Clear identification of missing, illegible and invalid values before analysis.
- Reusable definitions for recurring form types.
- Faster preparation of spreadsheet records and initial descriptive summaries.
- Improved ability for coordinators to review response patterns and data quality.
- A practical demonstration of responsible AI integration, where human review remains available.

These benefits are expected, not measured. A pilot should compare manual and assisted processing on equivalent batches.

18. LIMITATIONS AND FUTURE SCOPE

Extraction quality depends on handwriting, image quality, layout and AI-service availability. Schemas may need correction, and validation cannot detect every plausible but misread value. Local run instructions alone do not establish production readiness.

- Add rotation correction, deskewing, contrast adjustment and image-quality warnings.
- Build a verified benchmark and measure extraction accuracy and review effort.
- Prioritize review items by confidence, importance and error type.
- Strengthen multi-user access, storage isolation and monitoring before production deployment.
- Evaluate additional languages, multi-page forms and useful analytics extensions.

19. PROPOSED WORK PLAN

Phase	Activity	Deliverable
1	Define the problem and stakeholder workflow.	Requirements.
2	Review sample form and confirm schema.	Schema and constraints.
3	Run batch extraction and validation.	Structured records.
4	Review exceptions and test corrections.	Reviewed data.
5	Inspect workbook, analytics and report.	Exported outputs.
6	Run tests and document evaluation.	Synopsis and demonstration.

20. CONCLUSION

NIRMAN combines multimodal AI extraction, schema-driven processing, deterministic validation, explicit status handling, human review, Excel output and analytics in a reusable workflow. Its community value is the potential to make paper feedback easier to digitize, verify and summarize. AI output is checked against clear rules, with human review available for exceptions. Real-world accuracy, time savings, user acceptance and cost impact require formal evaluation.

REFERENCES

1. NIRMAN Data Automation Platform. Project README and technical documentation supplied for this synopsis.
2. Google AI for Developers. Gemini API documentation. ai.google.dev/gemini-api/docs.
3. Flask documentation. flask.palletsprojects.com.
4. Pydantic documentation. docs.pydantic.dev.
5. pytest documentation. docs.pytest.org.
6. Python documentation. docs.python.org/3/.

SUBMISSION / APPROVAL

Student 1	Student 2	Student 3	Project Guide
Uday Dhote	Tejas Jadhao	Vanshaj Dudhe	Dr. Pushpanjali Chouragade
Signature: _____	Signature: _____	Signature: _____	Signature: _____